Math 15 — Unit 4: Assignments

The assignment from every lesson in the booklet. Printed work is not tracked on the website.

Lesson 1 Assignment *Show your work.*

Part A — From Multiplying to Factoring

1. Each multiplication fact is done for you. Use it to write the matching factoring fact.

(a) Since $4(x+5) = 4x+20$, $4x+20 =$ _____

(b) Since $6(y-3) = 6y-18$, $6y-18 =$ _____

(c) Since $x(x+7) = x^2+7x$, $x^2+7x =$ _____

(d) Since $3x(x-2) = 3x^2-6x$, $3x^2-6x =$ _____

(e) Since $5(2x+1) = 10x+5$, $10x+5 =$ _____

(f) Since $2x(4x+3) = 8x^2+6x$, $8x^2+6x =$ _____

Part B — Greatest Common Factors

2. State the greatest common factor of each pair of numbers.

(a) 8, 12 (b) 15, 20 (c) 9, 21 (d) 14, 35 (e) 16, 24 (f) 18, 27

3. Write the prime factorization of $8x^2$ and $12x^3$, and use it to determine the greatest common factor of $8x^2$ and $12x^3$.

4. State the greatest common factor of each set of monomials by inspection.

(a) $6a,\ 9a^2$

(b) $10x^2,\ 15x$

(c) $4m^2n,\ 6mn^2$

(d) $12p^3,\ 18p^2$

(e) $7xy^2,\ 21x^2y$

(f) $8a^2b^2,\ 20ab^3$

Part C — Factoring and Checking

5. Complete the factoring in each case.

(a) $10x+15 =$ _____$(2x+3)$

(b) $6a^2-9a =$ _____$(2a-3)$

(c) $4xy+6x =$ _____$(2y+3)$

(d) $12m^2+8m+4 =$ _____$(3m^2+2m+1)$

6. Expand each proposed factored form to check it. If it does not match, write the correct factored form.

(a) Is $5x+20 = 5(x+4)$ correct?

(b) Is $7x+21 = 7(x+2)$ correct?

(c) Is $4x^2+12x = 4x(x+3)$ correct?

(d) Is $9x+6 = 3(3x+3)$ correct?

7. Factor each expression completely, and verify your answer by expanding it back out.

(a) $6x+9$ (b) $8x+20$ (c) $5x^2+15x$ (d) $9x^2+6x$

Part D — Applications, Errors and Review

8. The cost, in dollars, of manufacturing x items is given by $C = 5x + 25$.

(a) Factor C completely.

(b) Verify your factoring by expanding it back out.

(c) Use the factored form to find the cost when $x = 8$.

9. A student factored $6x + 8$ as $2(3x + 8)$. Identify the mistake, and write the correct factored form.

10. *(Multiple Choice)* Which expression shows $8x + 12$ factored completely, using the greatest common factor?

A. $4(2x + 3)$ B. $2(4x + 6)$ C. $4(2x + 8)$ D. $4x + 3$

11. *(Numerical Response)* When $12x+18$ is factored completely by removing the GCF, the result can be written as $a(bx + c)$. The value of $a + b + c$ is _____.

12. Factor each expression completely, and verify your answer by expanding it back out.

(a) $6x^2 + 10x$

(b) $15a^2b + 20ab^2$

(c) $9m^3 - 6m^2$

13. In your own words, explain why expanding the factored form is a good way to check that a factoring is correct. Then use this method to check whether $7x + 21 = 7(x + 2)$ is correct; if it is not, write the correct factored form.

Answer Key. 1. (a) $4(x + 5)$ (b) $6(y - 3)$ (c) $x(x + 7)$ (d) $3x(x - 2)$ (e) $5(2x + 1)$ (f) $2x(4x + 3)$. **2.** (a) 4 (b) 5 (c) 3 (d) 7 (e) 8 (f) 9. **3.** $8x^2 = 2\times2\times2\times x\times x$, $12x^3 = 2\times2\times3\times x\times x\times x$; GCF $= 4x^2$. **4.** (a) $3a$ (b) $5x$ (c) $2mn$ (d) $6p^2$ (e) $7xy$ (f) $4ab^2$. **5.** (a) 5 (b) $3a$ (c) $2x$ (d) 4. **6.** (a) correct (b) incorrect; $7(x + 3)$ (c) correct (d) incorrect; $3(3x + 2)$. **7.** (a) $3(2x + 3)$ (b) $4(2x + 5)$ (c) $5x(x + 3)$ (d) $3x(3x + 2)$. **8.** (a) $C = 5(x + 5)$ (b) $5(x + 5) = 5x + 25$ ✓ (c) \$65. **9.** Second term not divided by 2 ($8 \div 2 = 4$, not 8); correct form $2(3x + 4)$. **10.** A. **11.** $6 + 2 + 3 = 11$. **12.** (a) $2x(3x + 5)$ (b) $5ab(3a + 4b)$ (c) $3m^2(3m - 2)$. **13.** Expanding must reproduce the original exactly; $7(x + 2) = 7x + 14 \neq 7x + 21$, so incorrect — correct form $7(x + 3)$.

Lesson 2 Assignment *Show your work.*

Part A — Finding and Removing a GCF

1. State the greatest common factor of each set of monomials.
(a) $8x,\ 12x^2$ (b) $15a^3,\ 20a$ (c) $6xy,\ 9y$ (d) $-12m^2n,\ -18mn^2$

2. In each case the GCF has been removed. Complete the factoring.
(a) $9x + 15 =$ _____$(3x + 5)$ (b) $8a - 20 =$ _____$(2a - 5)$

(c) $6x^2 + 10x = 2x($_____$)$ (d) $12y - 18y^2 = 6y($_____$)$

3. Factor each binomial by removing the GCF. Verify your answer by expanding.
(a) $4x + 6$ (b) $10y - 15$ (c) $9a + 12$ (d) $21b - 14$ (e) $5x + 5$ (f) $16m - 24$

4. Factor each trinomial by removing the GCF.
(a) $2x^2 + 6x + 8$ (b) $5a^2 - 10a + 20$ (c) $9y^2 + 3y - 12$ (d) $6m^2 - 12m - 18$

Part B — GCFs That Include Variables

5. Factor each polynomial. The GCF includes a variable.
(a) $6x^3 + 9x^2$ (b) $10a^4 - 15a^3$ (c) $8y^3 + 4y^2$

(d) $12x^3 - 18x^2$ (e) $5m^3 + 15m^2$ (f) $4x^3 + 8x^2 + 16x$

6. Factor each polynomial. The GCF includes two variables.
(a) $8x^2y + 12xy^2$ (b) $10a^2b - 15ab^2$ (c) $6x^2y^2 + 9xy^3$

Part C — Negative GCFs

7. Factor each binomial by removing a *negative* GCF.
(a) $-3x - 9$ (b) $-6a + 18$ (c) $-5y - 20$ (d) $-8m + 12$

8. Factor each trinomial by removing a *negative* GCF.
(a) $-2x^2 + 6x - 4$ (b) $-3a^2 - 9a + 12$ (c) $-5y^2 + 10y + 15$

9. Factor each polynomial completely. The GCF is negative and includes a variable.
(a) $-4x^3 - 8x^2$ (b) $-6a^3 + 9a^2$ (c) $-12x^2y - 8xy^2$ (d) $-6m^3 + 12m^2 - 18m$

Part D — Applications and Review

10. A rectangular flower bed has area $A = 6x^2 + 9x$ (square metres), where x is a positive number of metres. (a) Factor A completely. (b) Use the factored form to find A when $x = 3$.

11. A student factored $-6x - 15$ as $-3(2x - 5)$. Identify the error in the student's work, and write the correct factored form.

12. *(Multiple Choice)* Following the convention in this lesson, what is the correct GCF to remove from $-8x^2 - 12x$? A. $4x$ B. $-4x$ C. 4 D. -4

13. *(Numerical Response)* When $-12x^3 + 18x^2$ is factored by removing the GCF, the result has the form $ax^2(bx + c)$. The value of $a \times b \times c$ is _____.

14. Factor each polynomial completely. Verify your answers by expanding.
(a) $8x^2 + 12x$ (b) $-5a^2 - 15a$ (c) $6x^3 - 9x^2$ (d) $-4y^3 + 8y^2 - 12y$

Answer Key. 1. (a) $4x$ (b) $5a$ (c) $3y$ (d) $-6mn$. **2.** (a) 3 (b) 4 (c) $3x + 5$ (d) $2 - 3y$. **3.** (a) $2(2x + 3)$ (b) $5(2y - 3)$ (c) $3(3a + 4)$ (d) $7(3b - 2)$ (e) $5(x + 1)$ (f) $8(2m - 3)$. **4.** (a) $2(x^2 + 3x + 4)$ (b) $5(a^2 - 2a + 4)$ (c) $3(3y^2 + y - 4)$ (d) $6(m^2 - 2m - 3)$. **5.** (a) $3x^2(2x + 3)$ (b) $5a^3(2a - 3)$ (c) $4y^2(2y + 1)$ (d) $6x^2(2x - 3)$ (e) $5m^2(m + 3)$ (f) $4x(x^2 + 2x + 4)$. **6.** (a) $4xy(2x + 3y)$ (b) $5ab(2a - 3b)$ (c) $3xy^2(2x + 3y)$. **7.** (a) $-3(x + 3)$ (b) $-6(a - 3)$ (c) $-5(y + 4)$ (d) $-4(2m - 3)$. **8.** (a) $-2(x^2 - 3x + 2)$ (b) $-3(a^2 + 3a - 4)$ (c) $-5(y^2 - 2y - 3)$. **9.** (a) $-4x^2(x + 2)$ (b) $-3a^2(2a - 3)$ (c) $-4xy(3x + 2y)$ (d) $-6m(m^2 - 2m + 3)$. **10.** (a) $A = 3x(2x + 3)$ (b) 81 m^2. **11.** The sign of the last term is wrong: $-3(2x - 5) = -6x + 15$, not $-6x - 15$. Correct form $-3(2x + 5)$. **12.** B. **13.** $(-6)(2)(-3) = 36$. **14.** (a) $4x(2x + 3)$ (b) $-5a(a + 3)$ (c) $3x^2(2x - 3)$ (d) $-4y(y^2 - 2y + 3)$.

Lesson 3 Assignment *Show your work.*

Part A — FOIL and the Sum–Product Pattern

1. Expand each product using FOIL.

(a) $(x+5)(x+6)$

(b) $(x+4)(x+4)$

2. Complete the table by finding two **positive** integers with the given sum and product.

Sum	Product	Integers
(a) 6	8	
(b) 8	12	
(c) 11	18	
(d) 12	32	

3. Complete the following.

(a) $x^2+8x+15$
$=(x+3)(x+___)$

(b) $x^2+10x+9$
$=(x+___)(x+___)$

(c) $x^2-10x+16$
$=(x-2)(x-___)$

(d) $t^2-13t+12$
$=(t-___)(t-___)$

(e) $z^2+14z+33$
$=(z+3)(_____)$

(f) $b^2-11b+18$
$=(b-2)(_____)$

Part B — Factoring x^2+bx+c

4. Factor each trinomial. (In each case, b and c are both positive.)

(a) x^2+6x+8

(b) $x^2+11x+10$

(c) $x^2+9x+14$

(d) $x^2+13x+12$

(e) $x^2+12x+20$

(f) $x^2+15x+26$

5. Factor each trinomial. (In each case, b is negative and c is positive.)

(a) x^2-6x+8

(b) $x^2-11x+10$

(c) $x^2-9x+14$

(d) $x^2-13x+12$

(e) $x^2-12x+20$

(f) $x^2-15x+26$

6. Factor where possible.

(a) $x^2+10x+16$

(b) $x^2-10x+16$

(c) $x^2+13x+22$

(d) $x^2-13x+22$

(e) $x^2+11x+15$

(f) x^2-8x+7

(g) $x^2+17x+16$

(h) $x^2-15x+36$

7. Factor completely. (Remove any common factor first.)

(a) $2x^2 + 16x + 24$ (b) $3x^2 - 33x + 90$ (c) $4x^2 + 44x + 72$ (d) $ax^2 + 10ax + 21a$

(e) $-2x^3 - 18x^2 - 40x$

Part C — Checking, Applications and Mixed Practice

8. Check each claimed factorization by expanding it with FOIL. State whether it is **correct** or **incorrect**; if incorrect, give the correct factorization.
(a) $x^2 + 12x + 27 \stackrel{?}{=} (x + 3)(x + 9)$ (b) $x^2 + 10x + 24 \stackrel{?}{=} (x + 4)(x + 7)$

9. A rectangular garden has an area of $x^2 + 12x + 35$ square metres.
(a) Factor $x^2 + 12x + 35$ to find binomials representing the length and width of the garden. (b) If $x = 4$, determine the length and width of the garden.

10. *(Multiple Choice)* The correct factorization of $x^2 + 15x + 36$ is
A. $(x + 3)(x + 12)$ B. $(x + 4)(x + 9)$
C. $(x + 6)(x + 6)$ D. $(x + 2)(x + 18)$

11. *(Multiple Choice)* Which of the following is **not** a factor of $2x^2 - 20x + 48$?
A. $x - 4$ B. $x - 6$ C. $x - 8$ D. 2

12. *(Numerical Response)* The trinomial $x^2 + bx + 24$ can be factored over the integers for several positive values of b. The **smallest** possible positive value of b is _____.

13. Factor each trinomial where possible.
(a) $x^2 + 8x + 7$ (b) $x^2 - 8x + 7$ (c) $x^2 + 17x + 30$ (d) $x^2 - 17x + 30$

(e) $x^2 + 19x + 18$ (f) $x^2 - 19x + 18$ (g) $x^2 + 10x + 25$ (h) $x^2 - 10x + 25$

14. Factor where possible.
(a) $x^2 + 11x + 28$ (b) $x^2 - 11x + 28$ (c) $x^2 + 10x + 21$ (d) $x^2 + 6x + 20$ (e) $x^2 - 13x + 36$

15. *(Multiple Choice)* The trinomial $x^2 - 9x + c$ **cannot** be factored (over the integers) if c has the value A. 14 B. 20 C. 18 D. 15

Answer Key. 1. (a) $x^2+11x+30$ (b) $x^2+8x+16$. **2.** (a) $2, 4$ (b) $2, 6$ (c) $2, 9$ (d) $4, 8$. **3.** (a) 5 (b) $1, 9$ (c) 8 (d) $1, 12$ (e) $(z+11)$ (f) $(b-9)$. **4.** (a) $(x+2)(x+4)$ (b) $(x+1)(x+10)$ (c) $(x+2)(x+7)$ (d) $(x+1)(x+12)$ (e) $(x+2)(x+10)$ (f) $(x+2)(x+13)$. **5.** (a) $(x-2)(x-4)$ (b) $(x-1)(x-10)$ (c) $(x-2)(x-7)$ (d) $(x-1)(x-12)$ (e) $(x-2)(x-10)$ (f) $(x-2)(x-13)$. **6.** (a) $(x+2)(x+8)$ (b) $(x-2)(x-8)$ (c) $(x+2)(x+11)$ (d) $(x-2)(x-11)$ (e) not possible (f) $(x-1)(x-7)$ (g) $(x+1)(x+16)$ (h) $(x-3)(x-12)$. **7.** (a) $2(x+2)(x+6)$ (b) $3(x-5)(x-6)$ (c) $4(x+2)(x+9)$ (d) $a(x+3)(x+7)$ (e) $-2x(x+4)(x+5)$. **8.** (a) Correct; $(x+3)(x+9) = x^2+12x+27$. (b) Incorrect; $(x+4)(x+7) = x^2+11x+28$, correct factorization is $(x+4)(x+6)$. **9.** (a) $(x+5)(x+7)$ (b) length 11 m, width 9 m. **10.** A. **11.** C. **12.** 10. **13.** (a) $(x+1)(x+7)$ (b) $(x-1)(x-7)$ (c) $(x+2)(x+15)$ (d) $(x-2)(x-15)$ (e) $(x+1)(x+18)$ (f) $(x-1)(x-18)$ (g) $(x+5)^2$ (h) $(x-5)^2$. **14.** (a) $(x+4)(x+7)$ (b) $(x-4)(x-7)$ (c) $(x+3)(x+7)$ (d) not possible (e) $(x-4)(x-9)$. **15.** D.

Lesson 4 Assignment *Show your work.*

Part A — Building the Opposite-Sign Pattern

1. Complete each table by finding two integers with the given sum and given product.

Sum	Product	Integers
(a) 11	-26	
(b) -11	-26	
(c) -4	-21	

Sum	Product	Integers
(d) 4	-21	
(e) -11	-12	
(f) 6	-16	

2. Complete the following.

(a) $x^2 + 5x - 14 = (x + 7)(x - ___)$

(b) $x^2 - 5x - 14 = (x - ___)(x + 2)$

(c) $n^2 + 3n - 28 = (n + ___)(n - ___)$

(d) $n^2 - 3n - 28 = (n - ___)(n + ___)$

(e) $y^2 + 2y - 35 = (y + 7)(_____)$

(f) $y^2 - 2y - 35 = (y - 7)(_____)$

Part B — Factoring with c Negative

3. Factor each trinomial.

(a) $x^2 + 4x - 5$

(b) $x^2 + x - 12$

(c) $x^2 + 3x - 28$

(d) $x^2 + 7x - 30$

(e) $x^2 + 2x - 8$

(f) $x^2 + 6x - 16$

4. Factor each trinomial.

(a) $x^2 - 4x - 5$

(b) $x^2 - x - 12$

(c) $x^2 - 3x - 28$

(d) $x^2 - 7x - 30$

(e) $x^2 - 2x - 8$

(f) $x^2 - 6x - 16$

5. Factor where possible.

(a) $x^2 + x - 20$

(b) $x^2 - 9x - 10$

(c) $x^2 + 3x - 40$

(d) $x^2 - 3x - 40$

(e) $x^2 + 5x - 6$

(f) $x^2 + 2x - 9$

(g) $x^2 - 4x - 32$

(h) $x^2 + 8x - 33$

Part C — Systematic Lists, Primes, and Common Factors

6. First list every factor pair of 48 and compute both possible sums (as in Example 3). Then use your table to factor each trinomial.

(a) $x^2 + 2x - 48$

(b) $x^2 - 2x - 48$

7. Decide whether each trinomial can be factored over the integers. If it can, factor it. If it cannot, write "prime."
(a) $x^2 + 5x - 9$ (b) $x^2 + 6x - 7$ (c) $x^2 + 3x - 11$ (d) $x^2 + 10x - 24$

8. Factor completely. Remember to remove any common factor first.
(a) $2x^2 + 2x - 40$ (b) $3x^2 - 3x - 36$ (c) $5x^2 + 15x - 50$

(d) $x^3 + 2x^2 - 15x$ (e) $2x^3 - 4x^2 - 48x$

Part D — Applications and Mixed Review

9. A rectangular garden has an area of $x^2 + 4x - 45$ square metres.
(a) Factor $x^2 + 4x - 45$ to find binomials representing the length and width of the garden.

(b) If $x = 12$, determine the length and width of the garden.

10. *(Multiple Choice)* Which of the following is **not** a factor of $3x^2 - 3x - 60$?
A. $x - 5$ B. $x + 4$ C. $x - 4$ D. 3

11. *(Multiple Choice)* For which trinomial is $(x - 6)$ **not** a factor?
A. $x^2 - x - 30$ B. $x^2 - 4x - 12$ C. $x^2 - 8x + 12$ D. $x^2 + x - 30$

12. *(Numerical Response)* The smallest **positive** value of b for which $x^2 + bx - 24$ can be factored over the integers is _____.

13. List all **positive** values of b for which $x^2 + bx - 18$ can be factored over the integers.

14. Factor where possible.

(a) $m^2 + 7m - 18$ (b) $m^2 - 7m - 18$ (c) $k^2 + 4k - 32$

(d) $k^2 - 4k - 32$ (e) $p^2 + p - 16$ (f) $p^2 + 6p - 27$

15. Factor completely.
(a) $4x^2 + 8x - 60$ (b) $6x^3 - 6x^2 - 72x$

Answer Key. 1. (a) $13, -2$ (b) $-13, 2$ (c) $-7, 3$ (d) $7, -3$ (e) $-12, 1$ (f) $8, -2$. **2.** (a) 2 (b) 7 (c) 7, 4 (d) 7, 4 (e) $(y - 5)$ (f) $(y + 5)$. **3.** (a) $(x+5)(x-1)$ (b) $(x+4)(x-3)$ (c) $(x+7)(x-4)$ (d) $(x+10)(x-3)$ (e) $(x+4)(x-2)$ (f) $(x+8)(x-2)$. **4.** (a) $(x-5)(x+1)$ (b) $(x-4)(x+3)$ (c) $(x-7)(x+4)$ (d) $(x-10)(x+3)$ (e) $(x-4)(x+2)$ (f) $(x-8)(x+2)$. **5.** (a) $(x+5)(x-4)$ (b) $(x-10)(x+1)$ (c) $(x+8)(x-5)$ (d) $(x-8)(x+5)$ (e) $(x+6)(x-1)$ (f) not possible (g) $(x-8)(x+4)$ (h) $(x+11)(x-3)$. **6.** (a) $(x+8)(x-6)$ (b) $(x-8)(x+6)$. **7.** (a) prime (b) $(x+7)(x-1)$ (c) prime (d) $(x+12)(x-2)$. **8.** (a) $2(x+5)(x-4)$ (b) $3(x-4)(x+3)$ (c) $5(x+5)(x-2)$ (d) $x(x+5)(x-3)$ (e) $2x(x-6)(x+4)$. **9.** (a) $(x+9)(x-5)$ (b) length 21 m, width 7 m. **10.** C. **11.** D. **12.** 2. **13.** 3, 7, 17. **14.** (a) $(m+9)(m-2)$ (b) $(m-9)(m+2)$ (c) $(k+8)(k-4)$ (d) $(k-8)(k+4)$ (e) not possible (f) $(p + 9)(p - 3)$. **15.** (a) $4(x + 5)(x - 3)$ (b) $6x(x - 4)(x + 3)$.

Lesson 5 Assignment *Show your work.*

Part A — Recognizing and Factoring

1. Complete each equation by determining the missing factor.
(a) $y^2 - 49 = (y - 7)(_____)$ (b) $h^2 - 144 = (h + 12)(_____)$ (c) $p^2 - q^2 = (p - q)(_____)$

2. Factor each difference of squares.
(a) $x^2 - 25$ (b) $x^2 - 4$ (c) $x^2 - 11^2$ (d) $x^2 - 400$

3. Explain why $x^2 - 64$ factors as $(x - 8)(x + 8)$ but $x^2 + 64$ cannot be factored this way.

4. Factor where possible.
(a) $a^2 - b^2$ (b) $g^2 - 36$ (c) $1 - k^2$ (d) $n^2 - 64m^2$

(e) $9x^2 - 25$ (f) $16a^2 - 9b^2$ (g) $4x^2 + y^2$ (h) $49c^2 - 100d^2$

Part B — Common Factors, Perfect Squares and Two Variables

5. The surface of a rectangular garden has area $9p^2 - 4q^2$ square metres.
(a) Write expressions for the length and width of the garden. (b) If the perimeter is 48 metres, form an equation in p and q and solve for p. (c) Determine the length and width if the length is twice the width.

6. Factor by first removing the GCF.
(a) $4x^2 - 16$ (b) $2a^2 - 50$ (c) $6t^2 + 54s^2$ (d) $5x^2 - 5y^2$

7. Factor completely.
(a) $3x^2y^2 - 27$ (b) $2m^3 - 32m$ (c) $45 - 5p^2q^2$

8. State whether each number is a perfect square. If it is, give its square root.
(a) 40 (b) 64 (c) 90 (d) 121 (e) 150 (f) 196

9. Factor completely.
(a) $x^2y^2 - 49$ (b) $a^2 - b^2c^2$ (c) $x^4 - y^2$ (d) $16m^2n^2 - 1$

Part C — Extensions

10. The pattern can multiply certain numbers mentally: $19 \times 21 = (20-1)(20+1) = 20^2 - 1^2 = 400 - 1 = 399$.
(a)(i) Evaluate 38×42. (a)(ii) Evaluate 59×61. (b) Explain why this process cannot be used, in the same simple way, for 22×27.

11. Decide whether each expression is a difference of squares. If it is, factor it completely; if not, explain why not.
(a) $x^2 - 81$ (b) $x^2 + 81$ (c) $9y^2 - 40$ (d) $49 - x^2$ (e) $16x^2 - 25y^4$

12. *(Multiple Choice)* One factor of $9x^2 - 16$ is
A. $3x - 4$ B. $9x - 4$ C. $3x + 16$ D. $x + 4$

13. *(Numerical Response)* $3x + 4y$ is a factor of the binomial $a^2x^2 - b^2y^2$. The value of $a + b$ is _____.

14. Factor completely: $12x^3 - 27xy^2$.

Answer Key. 1. (a) $(y+7)$ (b) $(h-12)$ (c) $(p+q)$. **2.** (a) $(x-5)(x+5)$ (b) $(x-2)(x+2)$ (c) $(x-11)(x+11)$ (d) $(x-20)(x+20)$. **3.** $64 = 8^2$, so $x^2 - 64$ is a true difference of squares; $x^2 + 64$ is a *sum* of squares — not factorable over the integers. **4.** (a) $(a-b)(a+b)$ (b) $(g-6)(g+6)$ (c) $(1-k)(1+k)$ (d) $(n-8m)(n+8m)$ (e) $(3x-5)(3x+5)$ (f) $(4a-3b)(4a+3b)$ (g) not factorable (sum) (h) $(7c-10d)(7c+10d)$. **5.** (a) $L = (3p+2q)$ m, $W = (3p-2q)$ m (b) $12p = 48$, $p = 4$ (c) $L = 16$ m, $W = 8$ m. **6.** (a) $4(x-2)(x+2)$ (b) $2(a-5)(a+5)$ (c) $6(t^2+9s^2)$, no further (d) $5(x-y)(x+y)$. **7.** (a) $3(xy-3)(xy+3)$ (b) $2m(m-4)(m+4)$ (c) $5(3-pq)(3+pq)$. **8.** (a) no (b) yes, 8 (c) no (d) yes, 11 (e) no (f) yes, 14. **9.** (a) $(xy-7)(xy+7)$ (b) $(a-bc)(a+bc)$ (c) $(x^2-y)(x^2+y)$ (d) $(4mn-1)(4mn+1)$. **10.** (a)(i) 1596 (ii) 3599 (b) the average of 22 and 27 is 24.5, not a whole number, so they are not the same whole number $\pm$ the same amount. **11.** (a) $(x-9)(x+9)$ (b) sum of squares — not factorable (c) 40 is not a perfect square — not factorable (d) $(7-x)(7+x)$ (e) $(4x-5y^2)(4x+5y^2)$. **12.** A. **13.** 7 ($a = 3$, $b = 4$). **14.** $3x(2x-3y)(2x+3y)$.

Lesson 6 Assignment *Show your work.*

Part A — Recognizing Perfect Square Trinomials

1. Use the three-part test to decide whether each trinomial is a perfect square trinomial.
(a) $x^2 + 4x + 4$ (b) $x^2 + 5x + 6$ (c) $x^2 - 12x + 36$ (d) $x^2 + 2x - 1$ (e) $4x^2 + 4x + 1$ (f) $x^2 - 10x + 16$

2. Fill in each blank so that the trinomial is a perfect square trinomial.
(a) $x^2 + ____ + 36$ (b) $x^2 - ____ + 49$ (c) $9x^2 + ____ + 16$ (d) $4x^2 + 20x + ____$

Part B — Factoring the Two Cases

3. Factor each perfect square trinomial (sum case).
(a) $x^2 + 12x + 36$ (b) $x^2 + 22x + 121$ (c) $x^2 + 2x + 1$ (d) $x^2 + 30x + 225$

4. Factor each perfect square trinomial (difference case).
(a) $x^2 - 12x + 36$ (b) $x^2 - 22x + 121$ (c) $x^2 - 2x + 1$ (d) $x^2 - 30x + 225$

5. Explain, using the three-part test, why $x^2 + 10x + 25$ is a perfect square trinomial but $x^2 + 10x + 24$ is not.

Part C — Choosing a Method

6. Decide whether each is a perfect square trinomial, then factor it completely, using the shortcut if it applies or the sum-product method if it does not.
(a) $x^2 + 20x + 64$ (b) $x^2 + 16x + 64$ (c) $x^2 - 13x + 40$ (d) $x^2 - 4x + 4$

7. Decide whether each is a perfect square trinomial, then factor it completely.
(a) $4x^2 + 28x + 49$ (b) $x^2 + 3x - 10$ (c) $9x^2 - 24x + 16$ (d) $x^2 - 4x - 21$

8. Factor completely by first removing the greatest common factor.
(a) $2x^2 + 20x + 50$ (b) $3x^2 - 42x + 147$ (c) $5x^2 + 60x + 180$

Part D — Application and Review

9. A square garden has area $(x^2 + 18x + 81)$ m^2.
(a) Write an expression for the side length of the garden. (b) If $x = 7$, find the numerical area and the numerical side length of the garden.

10. *(Multiple Choice)* Which of the following is **not** a perfect square trinomial?
A. $x^2 + 20x + 100$ B. $x^2 - 6x + 9$
C. $x^2 + 10x + 20$ D. $4x^2 + 12x + 9$

11. *(Multiple Choice)* The trinomial $x^2 - 24x + 144$ factors as
A. $(x - 12)^2$ B. $(x + 12)^2$
C. $(x - 12)(x + 12)$ D. $(x - 6)(x - 24)$

12. *(Multiple Choice)* One factor of $9x^2 + 30x + 25$ is
A. $3x + 5$ B. $9x + 5$ C. $3x + 25$ D. $x + 5$

13. *(Numerical Response)* The trinomial $16x^2 + Kx + 81$ is a perfect square trinomial, where K is positive. The value of K is _____.

14. Factor completely: $3x^3 + 30x^2 + 75x$.

Answer Key. 1. (a) yes, $(x + 2)^2$ (b) no (c) yes, $(x - 6)^2$ (d) no (e) yes, $(2x + 1)^2$ (f) no. **2.** (a) $12x$ (b) $14x$ (c) $24x$ (d) 25. **3.** (a) $(x + 6)^2$ (b) $(x + 11)^2$ (c) $(x + 1)^2$ (d) $(x + 15)^2$. **4.** (a) $(x - 6)^2$ (b) $(x - 11)^2$ (c) $(x - 1)^2$ (d) $(x - 15)^2$. **5.** $25 = 5^2$ and $2(x)(5) = 10x$, so $x^2 + 10x + 25 = (x + 5)^2$; but 24 is not a perfect square, so $x^2 + 10x + 24$ fails the test (it still factors as $(x + 4)(x + 6)$). **6.** (a) not PST, $(x + 4)(x + 16)$ (b) PST, $(x + 8)^2$ (c) not PST, $(x - 5)(x - 8)$ (d) PST, $(x - 2)^2$. **7.** (a) PST, $(2x + 7)^2$ (b) not PST, $(x + 5)(x - 2)$ (c) PST, $(3x - 4)^2$ (d) not PST, $(x + 3)(x - 7)$. **8.** (a) $2(x + 5)^2$ (b) $3(x - 7)^2$ (c) $5(x + 6)^2$. **9.** (a) $(x + 9)$ m (b) area 256 m^2, side 16 m. **10.** C. **11.** A. **12.** A. **13.** 72. **14.** $3x(x + 5)^2$.

Lesson 7 Assignment *Show your work.*

Part A — Finding the Two Integers

1. For each set of values, find $a \times c$, then find two integers with that product and with sum b.

	a	b	c	$a \times c$	**Integers**
(a)	2	9	4		
(b)	3	7	2		
(c)	2	-7	3		
(d)	4	1	-3		
(e)	3	-5	-2		
(f)	5	8	3		

2. Fill in the missing steps to factor $3x^2 + 11x + 6$ by decomposition.

$$3x^2 + 11x + 6 = 3x^2 + _____ + _____ + 6 = _____(x + 3) + 2(x + 3) = (x + 3)(_____)$$

Part B — Decomposition with Positive Coefficients

3. Factor each trinomial using decomposition.

(a) $2x^2 + 7x + 6$

(b) $3x^2 + 14x + 8$

(c) $4x^2 + 11x + 6$

(d) $2x^2 + 13x + 15$

4. Factor each trinomial using decomposition.

(a) $5x^2 + 17x + 6$

(b) $6x^2 + 13x + 5$

(c) $4x^2 + 16x + 15$

Part C — Negative b and/or c

5. Fill in the missing steps to factor $2x^2 - 7x + 3$ by decomposition.

$$2x^2 - 7x + 3 = 2x^2 - 6x - x + 3 = 2x(x - 3) - 1(_____) = (_____)(_____)$$

6. Factor each trinomial.

(a) $2x^2 - 11x + 5$

(b) $3x^2 - 13x + 4$

(c) $4x^2 - 19x + 12$

7. Factor each trinomial.

(a) $2x^2 + 3x - 5$

(b) $3x^2 - 2x - 8$

(c) $2x^2 + x - 6$

(d) $5x^2 - 9x - 2$

Part D — GCF First, Then Decompose

8. Factor completely. Remember to remove the GCF first.

(a) $2x^2 + 8x + 6$

(b) $4x^2 + 2x - 30$

(c) $9x^2 + 15x - 6$

(d) $10x^2 - 5x - 30$

9. Factor completely. The GCF includes a variable.

(a) $4x^3 + 10x^2 + 6x$

(b) $6x^3 - 3x^2 - 30x$

Part E — Applications and Review

10. A rectangular garden bed has area $3x^2 + 11x + 10$ m^2.

(a) Factor the area to find binomials representing the length and width of the bed.

(b) If $x = 4$, determine the length and width of the bed, in metres.

11. *(Multiple Choice)* One factor of $4x^2 + 4x - 24$ is

A. $x + 2$ B. $x - 2$ C. $2x + 3$ D. $2x - 3$

12. *(Multiple Choice)* The polynomials $2x^2 + x - 6$ and $3x^2 + 8x + 4$ have in common a factor of

A. $x + 2$ B. $x - 2$ C. $2x - 3$ D. $3x + 2$

13. *(Numerical Response)* The factored form of $4x^2 + 4x - 15$ is $(2x + a)(2x + b)$, where a and b are integers with $a > b$. The value of $a - b$ is _____.

14. *(Mixed Review)* Factor each expression completely, using whatever method applies.

(a) $6x + 15$

(b) $x^2 + 9x + 18$

(c) $x^2 - 3x - 18$

(d) $x^2 - 49$

(e) $3x^2 + 10x + 8$

(f) $6x^2 - 13x + 6$

Answer Key. 1. (a) $ac = 8$: $8, 1$ (b) $ac = 6$: $6, 1$ (c) $ac = 6$: $-6, -1$ (d) $ac = -12$: $4, -3$ (e) $ac = -6$: $-6, 1$ (f) $ac = 15$: $5, 3$. **2.** $9x$, $2x$; $3x$; $3x + 2$. **3.** (a) $(x + 2)(2x + 3)$ (b) $(x + 4)(3x + 2)$ (c) $(x + 2)(4x + 3)$ (d) $(x + 5)(2x + 3)$. **4.** (a) $(x + 3)(5x + 2)$ (b) $(3x + 5)(2x + 1)$ (c) $(2x + 5)(2x + 3)$. **5.** $(x - 3)$; $(x - 3)$, $(2x - 1)$. **6.** (a) $(x - 5)(2x - 1)$ (b) $(x - 4)(3x - 1)$ (c) $(x - 4)(4x - 3)$. **7.** (a) $(2x + 5)(x - 1)$ (b) $(x - 2)(3x + 4)$ (c) $(x + 2)(2x - 3)$ (d) $(x - 2)(5x + 1)$. **8.** (a) $2(x + 1)(x + 3)$ (b) $2(x + 3)(2x - 5)$ (c) $3(x + 2)(3x - 1)$ (d) $5(x - 2)(2x + 3)$. **9.** (a) $2x(2x + 3)(x + 1)$ (b) $3x(2x - 5)(x + 2)$. **10.** (a) $(x + 2)(3x + 5)$ (b) length 17 m, width 6 m. **11.** B. **12.** A. **13.** 8. **14.** (a) $3(2x + 5)$ (b) $(x + 3)(x + 6)$ (c) $(x - 6)(x + 3)$ (d) $(x - 7)(x + 7)$ (e) $(x + 2)(3x + 4)$ (f) $(2x - 3)(3x - 2)$.

Lesson 8 Assignment *Show your work.*

Part A — Naming the Strategy

1. Name the method (or methods, in order) you would use to factor each expression. You do **not** need to factor.

(a) $x^2 - 16$ (b) $x^2 + 5x + 6$ (c) $3x^2 + 12x$

(d) $2x^2 + 7x + 3$ (e) $x^2 + 8x + 16$ (f) $5x^2 - 45$

Part B — One Method at a Time

2. Factor completely (GCF only).

(a) $4x^2 + 8x$ (b) $6x^3 - 9x^2$ (c) $10xy - 15x$

3. Factor completely (difference of squares).

(a) $x^2 - 36$ (b) $4x^2 - 25$ (c) $9x^2 - 1$

4. Factor completely (trinomial, leading coefficient 1).

(a) $x^2 + 9x + 14$ (b) $x^2 - 4x - 21$ (c) $x^2 - 11x + 30$

Part C — Remove the GCF First

5. Factor completely. (Hint: remove the GCF first.)

(a) $2x^2 + 16x + 24$ (b) $3x^2 - 3x - 36$ (c) $5x^2 + 25x + 30$

6. Factor completely. (Hint: this will take two steps.)

(a) $3x^2 - 27$ (b) $2x^2 - 72$ (c) $7x^2 - 63$

Part D — Perfect Squares and Harder Trinomials

7. Factor completely (perfect square trinomials).

(a) $x^2 + 6x + 9$ (b) $x^2 - 14x + 49$ (c) $4x^2 + 12x + 9$

8. Factor completely (trinomial, leading coefficient not 1).

(a) $2x^2 + 11x + 5$ (b) $3x^2 - 10x + 8$ (c) $2x^2 + x - 15$

Part E — Mixed Practice and Applications

9. Factor completely. Use the checklist to decide which method applies each time.
(a) $x^2 - 81$ (b) $2x^2 + 18x + 40$ (c) $x^2 + 12x + 36$

(d) $3x^2 - 48$ (e) $x^2 - 2x - 15$ (f) $5x^2 + 13x - 6$

10. Factor completely.
(a) $4x^2 - 4x - 24$ (b) $6x^2 + 42x + 72$

11. A rectangular picture frame has area $x^2 + 3x - 40$ square inches.
(a) Factor the area to find binomials for the length and width. (b) If the width is $x - 5$, state the length. (c) If $x = 12$, determine the actual length and width, in inches.

12. *(Multiple Choice)* $3x^2 - 12$ is completely factored as
A. $3(x - 2)(x + 2)$ B. $3(x^2 - 4)$
C. $(3x - 6)(x + 2)$ D. $x(3x - 12)$

13. *(Numerical Response)* If $x^2+kx+64$ is a perfect square trinomial and $k > 0$, then the value of k is _____.

14. Factor completely: $8x^2 + 40x + 48$.

15. Factor completely: $2x^3 - 8x$.

Answer Key. 1. (a) difference of squares (b) trinomial, $a = 1$ (inspection) (c) GCF only (d) trinomial, $a \neq 1$ (decomposition) (e) perfect square trinomial (f) GCF, then difference of squares. **2.** (a) $4x(x + 2)$ (b) $3x^2(2x - 3)$ (c) $5x(2y - 3)$. **3.** (a) $(x - 6)(x + 6)$ (b) $(2x - 5)(2x + 5)$ (c) $(3x - 1)(3x + 1)$. **4.** (a) $(x + 7)(x + 2)$ (b) $(x - 7)(x + 3)$ (c) $(x - 5)(x - 6)$. **5.** (a) $2(x + 6)(x + 2)$ (b) $3(x - 4)(x + 3)$ (c) $5(x + 2)(x + 3)$. **6.** GCF then DoS: (a) $3(x - 3)(x + 3)$ (b) $2(x - 6)(x + 6)$ (c) $7(x - 3)(x + 3)$. **7.** (a) $(x + 3)^2$ (b) $(x - 7)^2$ (c) $(2x + 3)^2$. **8.** (a) $(x + 5)(2x + 1)$ (b) $(x - 2)(3x - 4)$ (c) $(x + 3)(2x - 5)$. **9.** (a) $(x - 9)(x + 9)$ (b) $2(x + 5)(x + 4)$ (c) $(x + 6)^2$ (d) $3(x - 4)(x + 4)$ (e) $(x - 5)(x + 3)$ (f) $(x + 3)(5x - 2)$. **10.** (a) $4(x - 3)(x + 2)$ (b) $6(x + 3)(x + 4)$. **11.** (a) $(x + 8)(x - 5)$ (b) length $= x + 8$ (c) length 20 in, width 7 in. **12.** A. **13.** 16 (since $(x + 8)^2$ needs $k = 2 \times 8$). **14.** GCF then trinomial: $8(x + 2)(x + 3)$. **15.** GCF then DoS: $2x(x - 2)(x + 2)$.

Lesson 9 Assignment *Show your work.*

Part A — The Zero Product Property

1. Solve each equation using the Zero Product Property.

(a) $(x-3)(x+8)=0$ (b) $(x+5)(x-1)=0$ (c) $(x+10)(x+2)=0$

(d) $6(x-7)(x+4)=0$ (e) $x(x-11)=0$ (f) $3x(x+5)=0$

2. Solve each equation by factoring.

(a) $x^2-36=0$ (b) $x^2+10x=0$ (c) $x^2-8x+15=0$

(d) $x^2+x-20=0$ (e) $x^2-6x=0$ (f) $x^2-81=0$

Part B — Rearranging Before Factoring

3. Solve each equation. Rearrange first so one side is 0.

(a) $x^2+7x=8$ (b) $x^2=5x+6$

(c) $x^2-2x=15$ (d) $x^2+4x=21$

4. Solve each equation, then check both solutions by substituting into the original equation.

(a) $x^2-2x=8$ (b) $x^2+x=6$

Part C — Decomposition and Mixed Practice

5. Solve each equation using decomposition (leading coefficient not equal to 1).

(a) $2x^2+7x+3=0$ (b) $3x^2-5x-2=0$ (c) $2x^2-x-6=0$

6. Explain why you **cannot** solve $(x-3)(x+9)=13$ by simply setting $x-3=13$ and $x+9=13$. Then solve the equation correctly.

7. Solve: $x^2 - 3x = 0$

8. Solve: $2x^2 + 3x - 2 = 0$

9. Solve $x^2 + 6x = 16$, then verify both roots by substitution.

10. Solve: $4x^2 - 25 = 0$

11. Solve: $3x^2 + 11x + 6 = 0$

12. Solve: $x^2 = 7x - 10$

Part D — Multiple Choice and Numerical Response

13. *(Multiple Choice)* The complete solution to the equation $x(x-3) = 10$ is A. $x = 5$ and $x = 2$ B. $x = -5$ and $x = 2$ C. $x = 5$ and $x = -2$ D. $x = -5$ and $x = -2$

14. *(Numerical Response)* The equation $x^2 + kx - 24 = 0$ has one solution $x = -8$. The value of k is _____.

15. *(Numerical Response)* The positive solution of $6x^2 - 7x - 3 = 0$ can be written as $x = \frac{p}{q}$ in lowest terms, where p and q are positive integers. The value of $p + q$ is _____.

Answer Key. 1. (a) $x = 3, -8$ (b) $-5, 1$ (c) $-10, -2$ (d) $7, -4$ (e) $0, 11$ (f) $0, -5$. **2.** (a) ± 6 (b) $0, -10$ (c) $3, 5$ (d) $-5, 4$ (e) $0, 6$ (f) ± 9. **3.** (a) $-8, 1$ (b) $6, -1$ (c) $5, -3$ (d) $-7, 3$. **4.** (a) $4, -2$ (b) $-3, 2$ (both check). **5.** (a) $-3, -\frac{1}{2}$ (b) $2, -\frac{1}{3}$ (c) $2, -\frac{3}{2}$. **6.** The zero product property applies only when a product equals 0, not 13; expand and rearrange so one side is 0 first: $x^2 + 6x - 40 = 0$, so $x = -10, 4$. **7.** $0, 3$. **8.** $-2, \frac{1}{2}$. **9.** $-8, 2$ (both check). **10.** $\pm\frac{5}{2}$. **11.** $-3, -\frac{2}{3}$. **12.** $5, 2$. **13.** C. **14.** 5. **15.** 5 (since $x = \frac{3}{2}$).

Lesson 10 Assignment *Show your work.*

Part A — Removing a Greatest Common Factor

Factor each expression completely.

1. $14x^2 - 21x$
2. $-8x^3 + 20x^2 - 12x$
3. $24x^2y^3 - 16x^3y^2$

Part B — Trinomials $x^2 + bx + c$

Factor each trinomial. (Questions 4–5 have c positive; questions 6–7 have c negative.)

4. $x^2 + 14x + 45$
5. $x^2 - 16x + 63$
6. $x^2 + 3x - 70$
7. $x^2 + 2x - 63$

Part C — Difference of Squares and Perfect Square Trinomials

Factor each expression.

8. $x^2 - 169$
9. $25x^2 - 16$
10. $x^2 + 38x + 361$
11. $9x^2 - 42x + 49$

Part D — Decomposition ($a \neq 1$)

Factor each trinomial using decomposition.

12. $4x^2 + 13x + 3$
13. $5x^2 - 14x + 8$
14. $6x^2 + 7x - 20$

Part E — Choosing a Strategy

Factor each expression completely. Remember to look for a GCF first.

15. $3x^3 - 3x$
16. $2x^2 + 8x + 8$
17. $4x^3 + 20x^2 - 96x$
18. $3x^2 - 108$

Part F — Solving Equations by Factoring

Solve each equation by factoring.

19. $x^2 + 9x - 52 = 0$
20. $2x^2 - 5x - 12 = 0$
21. $16x^2 = 49$
22. $x^2 + 7x = 18$

Part G — Numerical Response and Written Response

23. *(Numerical Response)* When $3x^3 - 12x$ is factored completely, it can be written in the form $ax(x-b)(x+c)$, where a, b, and c are positive integers. The value of $a+b+c$ is _____.

24. *(Written Response)* Two students, Jamal and Priya, attempt to factor $x^2 + 5x - 66$. Their work is shown below. Each student's work contains exactly one mathematical error.

Jamal	Priya
$x^2 + 5x - 66$	$x^2 + 5x - 66$
Factors of 66 differing by 5: 11 and 6.	Factors of 66 differing by 5: 11 and 6.
$(x+6)(x-11)$	$(x+11)(x+6)$

(a) Describe in detail the error Jamal made.
(b) Describe in detail the error Priya made.
(c) Determine the correct factored form of $x^2 + 5x - 66$.
(d) Verify your factoring by substituting $x = 3$ into both the original expression and your factored form, confirming the same numerical value.

Answer Key. **1.** $7x(2x-3)$ **2.** $-4x(2x-3)(x-1)$ **3.** $8x^2y^2(3y-2x)$ **4.** $(x+5)(x+9)$ **5.** $(x-7)(x-9)$ **6.** $(x+10)(x-7)$ **7.** $(x+9)(x-7)$ **8.** $(x-13)(x+13)$ **9.** $(5x-4)(5x+4)$ **10.** $(x+19)^2$ **11.** $(3x-7)^2$ **12.** $(4x+1)(x+3)$ **13.** $(5x-4)(x-2)$ **14.** $(3x-4)(2x+5)$ **15.** $3x(x-1)(x+1)$ **16.** $2(x+2)^2$ **17.** $4x(x+8)(x-3)$ **18.** $3(x-6)(x+6)$ **19.** $x = -13, 4$ **20.** $x = -\frac{3}{2}, 4$ **21.** $x = \pm\frac{7}{4}$ **22.** $x = -9, 2$ **23.** 7 **24.** Jamal: right numbers, wrong signs — $+11$ and -6 are needed since $b = +5$; Priya: both signs positive, giving product $+66$ not -66. Correct: $(x+11)(x-6)$; at $x = 3$ both give -42.